A modern design approach at Mtkvari, Georgia

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The Mtkvari hydro plant is under construction on the Mtkvari river in Georgia. This scheme has a strategic importance for the development of a stable, reliable, cost-effective and efficient national electricity supply. This article looks at the main features of this project, as well as at the technical solutions that have been taken in the interest of safety, to comply with good international practice.

While the last decade has seen the hydropower sector contribute almost 79 per cent of Georgia's electricity generation, the government's energy strategy is to double its hydropower capacity over the next decade to increase energy security, develop cross-border electricity trade opportunities, and reduce reliance on fossil fuel imports by tapping the still significant unexploited hydropower potential of the country.

One of the major ongoing hydropower schemes in Georgia is the 53 MW Mtkvari project developed by Mtkvari Hesi LLC, which is expected to produce 260 GWh/yr. The project is located on the Mtkvari river in the Aspindza and Akhaltsikhe districts of the Samtskhe-Javakheti region in Southern Georgia, close to the Turkish border.

1. Overview

1.1 Project background

The project was launched in 2009. It was then purchased in 2014 by the Georgian Co-Investment Fund, and the construction works restarted in 2015. In 2021, Gruner Stucky was appointed as the owner's engineer, responsible for construction supervision activities, design review and technical assistance. One of the main tasks was the review of the initial design concept, including the structures already built, and to define optimisation measures to allow for the design to comply with good international practice.

1.2 Headworks

The headworks of Mtkvari feature an embankment dam with its crest at el. 1019.74. Maximum flood level, full supply level and minimum operating level for the Mtkvari reservoir are, respectively, el. 1018.04, el. 1016.74 and el. 1015.74. The reservoir allows for a total storage of $6.13 \times 10^6 m^3$, with a live storage capacity of $0.53 \times 10^6 m^3$ and a flood storage capacity of $0.93 \times 10^6 m^3$. The dam is divided into four sections, creating two reservoir areas.

- *Dam S-1*: A 25 m-high and 219 m-long embankment dam blocks the first bend sleeve on the river. It is constructed as a permeable embankment made of gravel and pebbles fill and acts as a bridge between dam S-2 and the existing road. It divides the reservoir into two areas, which are connected by a culvert.
- *Dam S-2*: A 28 m-high and 374 m-long embankment blocks the second bend sleeve on the river and constitutes the main water retaining structure.
- *Dam S-3*: A 5 m-high and 83 m-long embankment connects dam S-2 with the existing appurtenant structures. It is founded on rock and includes a retaining wall acting as a watertight element.
- *Dam S-4*: A 9 m-high lateral embankment with clay fill develops over the natural ridge and it is connected to the fish pass by a retaining wall.

The existing appurtenant structures include an ungated surface spillway and a river bottom outlet located between dam S-3 and S-4. The bottom outlet is 127.1 m-long, featuring two openings of 6 × 6 m each, equipped with two gates that can be operated only by a gantry crane. It is planned to upgrade the two bottom outlet gates by installing a self-lifting mechanism for each gate, to provide more flexibility during impounding and operation. The ungated surface spillway is located above the bottom outlet and features a box-shaped layout. Its intakes are located on three sides of the structure at el. 1016.74, with a total crest length of 46 m. Following the assessment completed by Gruner Stucky, it was deemed necessary to include a new main service spillway within dam S-4, located on the left-hand side of the existing outlet works, to optimise the safe discharge of the design flood and to handle the safety check flood. A small hydropower plant was also planned to be included within the structure of the new main spillway to use the environmental flow discharged from the dam to produce additional electricity.

1.3 Waterways

The waterways of the scheme feature an intake structure, a 9.6 km-long TBM-driven headrace tunnel, and a 42.5 m-deep surge tank with a diameter of 19.5 m, which is connected to the top of the headrace tunnel by a 4.5 m-diameter vertical shaft. The headrace tunnel bypasses an approximately 27 km-long reach of the Mtkvari river. The headrace tunnel was originally lined with 5.26 m-diameter precast concrete elements. However, following its construction, remedial lining was carried out in nine sections with total length of 1536 m by placing 0.25 m-thick additional reinforced concrete. The last 270 m of the headrace tunnel is steel-lined, and it is followed by a surface bifurcation structure and two penstocks embedded in concrete. Each penstock is equipped with an emergency valve.

1.4 Powerhouse

The powerhouse is equipped with two Francis turbines with a rated capacity of 27.04 MW each, and the switchyard is located on the immediate downstream of the powerhouse. The gross head of the scheme can vary from 103.3 to 105.22 m, and the rated discharge is 32.92 m^3/s per unit.

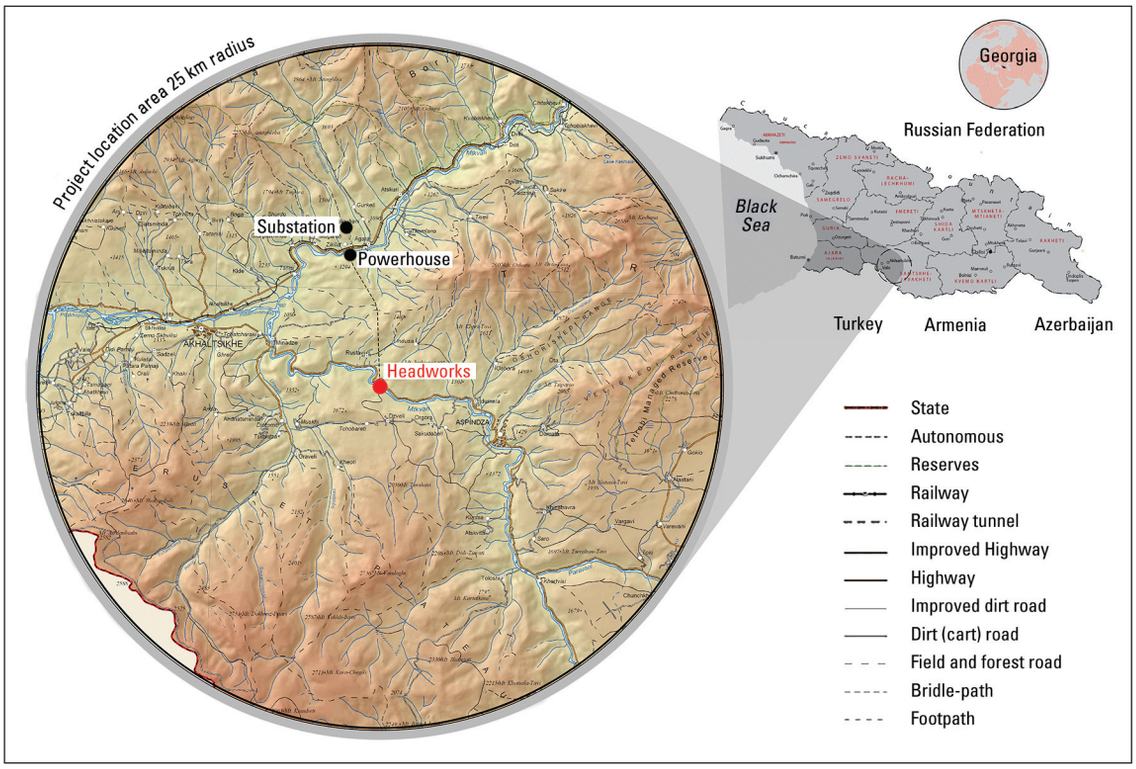


Fig. 1. The location of the Mtkvari project in Georgia.

2. Project design optimization

2.1 Dam S-2

Dam S-2 was built up to about el. 1005 as an alluvial-fill embankment dam with a central clay core and with a cut-off wall reaching the bedrock. From el. 1005, the design of dam S-2 was optimised by Gruner Stucky as a geomembrane-faced embankment dam by considering these aspects:

- Installation of a geomembrane on the upstream face of dam S-2 as a water barrier. Above el. 1004, the geomembrane is exposed, while below el. 1004 it is covered by a ballast fill. The geomembrane is anchored to the dam crest at el. 1019.04.
- Construction of a reinforced concrete plinth at the upstream toe of the dam embankment, as well as on the left and right banks, where the geomembrane is connected.
- A closing reinforced concrete wall is being constructed 0.5 m above and along the left side of dam S-2. It is founded on sound rock to ensure watertightness at the interface between dams S-2, S-1 and S-3. The geomembrane is designed to be connected to the crest of this wall.
- A horizontal drainage layer crossing the dam body from upstream to downstream is being built above the clay core, to provide a preferred drainage path within the embankment.
- Instrumentation such as leakage detection optical fibres behind the geomembrane, pore pressure cells, standpipe piezometers within the dam body and foundation, as well as a geodetic network, are planned to be installed to monitor the behaviour of the dam for impounding, and during its operational life.

Static and dynamic global stability, as well as stress-strain analysis, were conducted to evaluate the behaviour of the final design solution for dam S-2 under different loads and combinations of them. This was carried out based on good engineering practice, according to recognized industry guidance (for example, USACE and USBR).

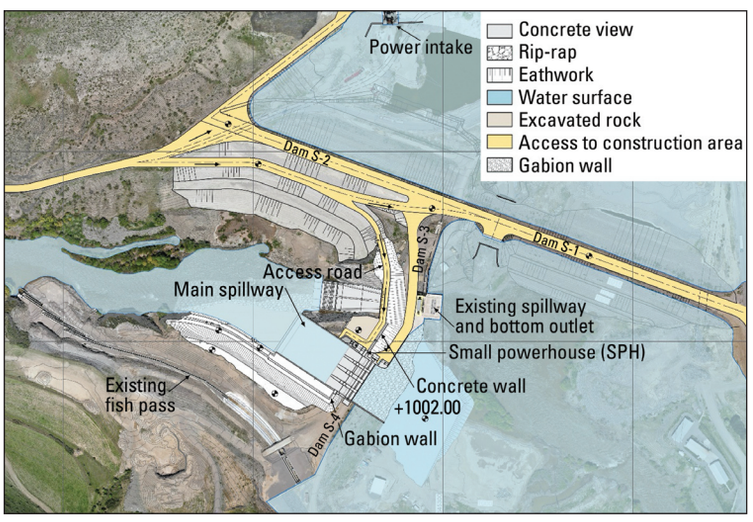


Fig. 2. General arrangement of the Mtkvari headworks.



The clay core of Dam S-2.

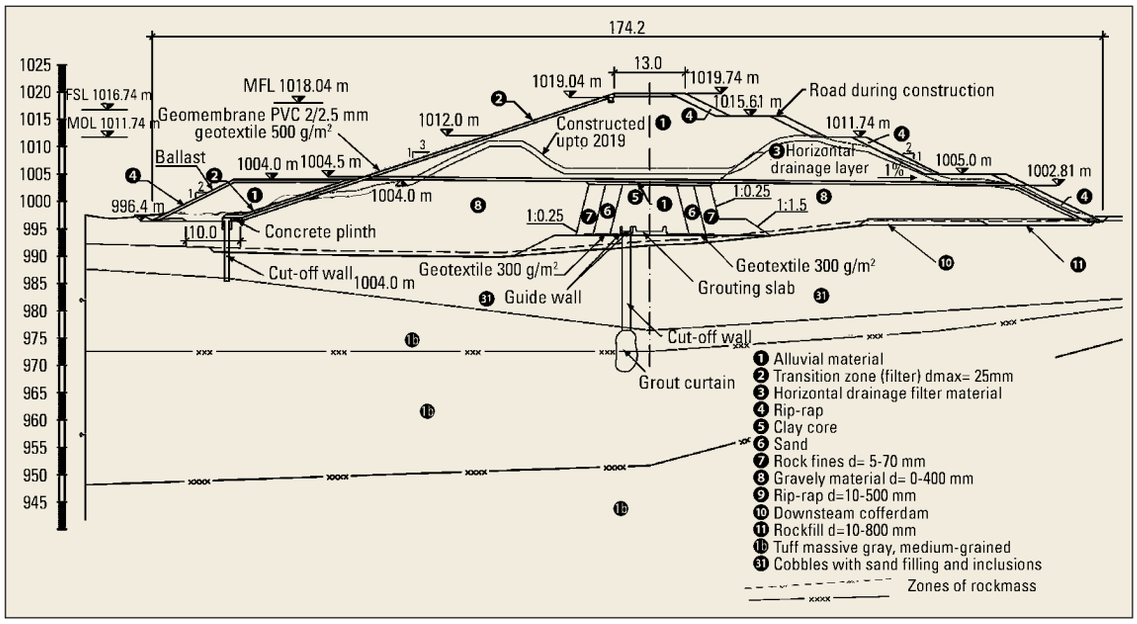


Fig. 4. Final design of the typical section of dam S-2.

2.2 Main spillway

The new main spillway was designed according to ICOLD Bulletin 170, based on a design flood with a return period of 10 000 years (1900 m^3/s), assumed to be released with the gate of highest capacity not being available, and a safety check flood equal to the PMF (3400 m^3/s). With the selection of these design criteria, and following a study, the main spillway location, type, and size were optimized by Gruner Stucky based on a technical and economic appraisal, according to recognized industry guidance (such as, USACE and USBR). The main spillway is planned to be built within dam S-4 on the left-hand side of the existing outlet works. It consists of a gated overflow ogee crest control structure with an upstream approach channel, and a downstream chute followed by a stilling basin and exit channel. Main features of its design layout are described as follows:

- *Approach channel*: Flow is conveyed in a rectangular channel with riprapped-lined invert.
- *Control structure*: This consists of an overflow ogee crest divided into four 9 m-wide bays, each equipped with a radial gate. The ogee crest is at el. 1007.25. A grouting and drainage gallery is located within the ogee structure.
- *Terminal structure*: A reinforced concrete rectangular stilling basin with a horizontal floor and sloping end sill is located at the end of the spillway chute, to dissipate the energy of the flow and to protect the downstream area from erosion and scouring.
- *Exit channel*: an excavated exit channel to provide a transition between the stilling basin and the main channel of the downstream Mtkvari river. Riprap protection is extended downstream from the channel.

The main spillway is currently undergoing 3D numerical hydraulic (CFD) model testing, and structural and geotechnical detailed design.

2.3 Small hydro powerhouse

To allow for the production of additional electricity, it was planned to install a small hydro plant, to take advantage of the civil works required for the main spillway, by using the 5.8 m^3/s constant outflow required to be released from the headworks of Mtkvari as the environmental flow. After turbining, thed flow will then be then released into the stilling basin of the main spillway. Based on the available flow, head and site configuration, a single 1.1 MW vertical Saxo turbine was recommended. The powerhouse is designed as a shaft-type structure integrated within the right-

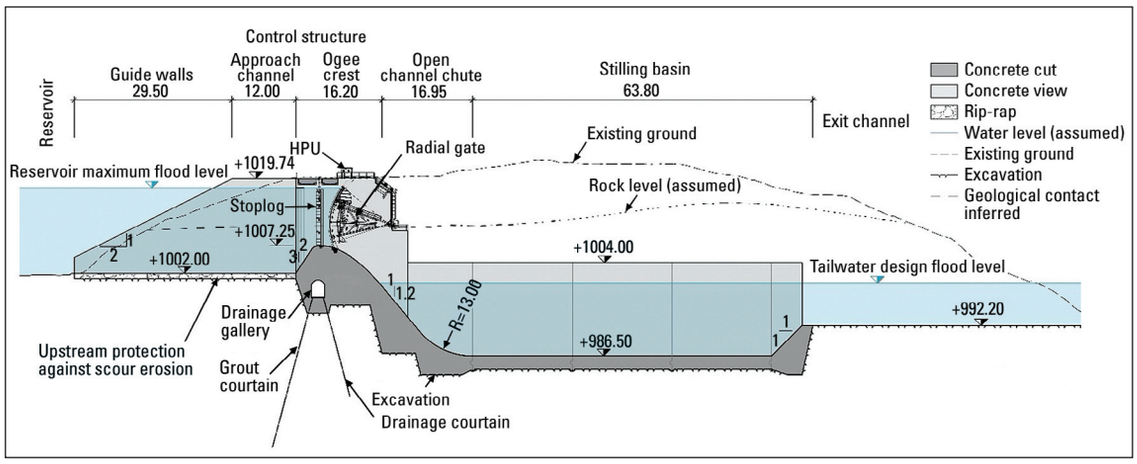


Fig. 5. Longitudinal profile of the main spillway.



Fig. 6. Site of the Mtkvari headworks (in October 2022).



Fig. 7. The Mtkvari powerhouse construction site (in October 2022).

hand sidewall of the control structure of the main spillway. Some design aspects are still under review, and the final solution has not yet been fixed.

3. Construction activities

The present status of the project is summarized below. Gruner Stucky is supervising the remaining construction activities and installation of equipment on site. Commissioning of Mtkvari is planned for the beginning of 2024.

- *Relocation of access road*: from the headworks towards upstream, the public road E691 needed to be relocated uphill for a length of about 2.45 km for the impounding of the reservoir. The main works regarding this new stretch of road have been completed.
- *Headworks*: for dam S-2, the excavation works for the removal of the protective layer over the existing clay core and stripping on the upstream and downstream slopes as well as backfilling of clay core have been completed. Trial embankment filling was initiated in August 2022. Construction of the closing wall foundation at the left-hand side of dam S-2 is ongoing. Bulk excavation for the new main spillway has started in August 2022 and is ongoing. Structure of the power intake at the inlet of the headrace tunnel is completed.
- *Headrace tunnel*: excavation of the tunnel by mean of a TBM is completed, as well as the installation of precast concrete elements. Works related to the installation of remedial concrete linings are ongoing.
- *Surge facility*: works to raise the top elevation of the surge tank by building a new concrete wall above the existing ground level have been initiated. Installation of the T-junction and vertical steel lining at the lower connection with the tunnel and within the surge shaft is completed.
- *Steel lining*: installation of the steel lining within the last 270 m of the tunnel is completed.
- *Bifurcation, emergency valves and penstocks*: the penstock installation works are ongoing, while the bifurcation and valves are in the design and procurement phase.
- *Powerhouse*: main structures of the powerhouse building are completed. The main electromechanical equipment, including two generating units, spiral cases, inlet valves and last section of the penstocks was installed, as well as the two transformers. Installation of auxiliary electrical equipment and mechanical works, as well as architectural finishing, is ongoing. Connection of the tailrace channel to Mtkvari river is planned to be completed during the low flow season, before the commissioning of the plant.
- Switchyard: civil works and installation of electrical equipment for the switchyard are completed.
- Overhead line: activities regarding the implementation of the overhead line between the switchyard and the dispatch centre, including relocation of some existing towers, are under finalisation process.

4. Conclusion

As a result of an open and honest collaboration between the Client, Mtkvari Hesi LLC, and the owner's engineer, Gruner Stucky, the original design of the project was optimised by means of cost-effective and state-of-the-art technical solutions. This will ensure the sustainable production of renewable energy by means of a safe and reliable power asset, contributing to satisfy the increasing energy demand of the region. ◊



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